

Option Pricing, Computational Finance and Machine Learning

Lesson 1

Option Pricing: a Practitioner Approach

Antoine Savine

NYU

September 16, 2026

A Practitioner Approach to Option Pricing

- "Practitioner Approach" does *not* mean that we investigate gritty practical details – in fact, this is the opposite: we assume idealized markets without rates or dividends and focus on the core phenomena at play when pricing and hedging options
- What we mean by "practitioner approach" is that we investigate the profits and losses that option traders experience in the real world when trading options, and *from there*, re-derive the main results of the option pricing theory, including:
 - ▶ Black and Scholes' PDE
 - ▶ Risk-neutral pricing
 - ▶ Bachelier, Black and Scholes and Dupire formulas
 - ▶ The uber-important *mis-replication formula*
- In the next lesson, we take a "theoretical approach", where we generalize and abstract those results so as to encompass:
 - ▶ All kind of Derivatives instruments, not only European options
 - ▶ Multi-underlying models and instruments
 - ▶ "Exotic" underlying markets, such as interest rates

Outline

- 1 Option Pricing
- 2 Option Hedging and PnL
- 3 Black and Scholes PDE
- 4 Volatility Models
- 5 Risk-Neutral Pricing
- 6 Constant Volatility Models
- 7 Mis-Replication Formulas
- 8 Local Volatility Models

Option Pricing

European Call

Definition (European Call)

Contract that provides the *holder* the right (but not the obligation) to purchase some *underlying* asset on a future date (called *expiry* or *maturity* and denoted T) for a fixed price (called *strike* and denoted K)

Definition (Payoff)

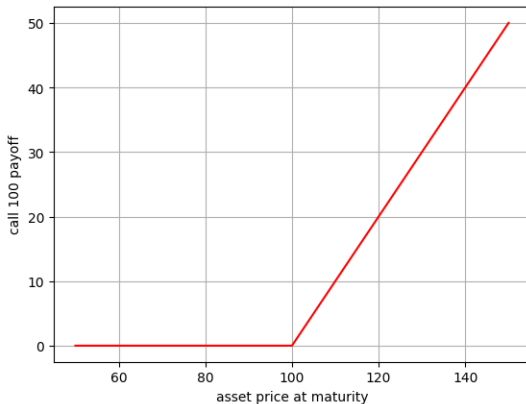
The *payoff* of the call is the (unknown) profit that the holder realizes at maturity

- Denote S_t the price of the underlying asset at time t – the (known) price today is S_0 – the (random) price at maturity is S_T
- If $S_T \leq K$ the holder has no interest to *exercise* the call. The option expires *out of the money*, with payoff 0
- If $S_T > K$ the holder exercises the option and purchases for K an asset worth S_T , realizing a profit of $S_T - K$
- Hence, the payoff of a call (which must coincide with its price at maturity T):

$$C(S_T, T) = \max(0, S_T - K) = (S_T - K)^+$$

European Call: Payoff

$$C(S_T, T) = \max(0, S_T - K) = (S_T - K)^+$$



European Call: Pricing and Hedging

- The European call is the simplest example of a nonlinear Derivatives contract
 - Derivatives contracts are bilateral agreements that promise the exchange of cash-flows depending of the future behavior of some underlying assets
 - European calls deliver a payoff that depends on the market price S_T of some underlying asset at maturity T
 - The payoff is a nonlinear function of S_T
 - European calls are massively traded on many markets under different forms
-
- The question of *pricing* and *hedging* European calls was addressed by Black and Scholes in a ground-breaking article dated 1973
 - Black and Scholes' foundational article also pioneered stochastic calculus in finance, and founded modern Derivatives theory
 - In this lesson, we are going to investigate these questions from the perspective of option traders and re-derive those results from an analysis of their profits and losses (PnL)

Assumptions

No arbitrage

No trading strategy can generate all positive cash-flows with probability 1, or infinite cash-flows with probability > 0

For the entire course

Continuous paths underlying asset price series are continuous

Regularity conditions price series are regular enough to exclude degenerate edge cases where calculations are not well-defined

Idealized markets no transaction costs, infinite liquidity, transparent and costless short-selling

Default-free markets no credit risk

No dividends

For the first few lessons only: no interest rates

Interest rates are and stay 0

European Call Pricing: First Properties

Notation

$C(K, T)$ denotes the current price of the call with strike K and maturity T

Trivial Properties

- $C(K, T) \geq 0$ since holders have no obligations, only rights
- When $S_0 \gg K$, the option will be exercised, its price must go to $S_0 - K$
- When $S_0 \ll K$, the option will not be exercised, its price converges to 0
- Call prices must decrease with strike:

$$\frac{\partial C}{\partial K} \leq 0$$

Why?

European Call Pricing: First Properties

Less Trivial Properties

- A call is always worth at least its *intrinsic value* $(S_0 - K)^+$
 - ▶ **Why?**
 - ▶ The positive difference $C(K, T) - (S_0 - K)^+$ is called *time value*
 - ▶ (Obviously) time value converges to 0 at maturity (when T goes to 0)

- Call prices must increase with maturity:

$$\frac{\partial C}{\partial T} \geq 0$$

Why?

- Call prices must be *convex* in strike:

$$\frac{\partial^2 C}{\partial K^2} \geq 0$$

Why?

European Call: Pricing Function

The price C_t of a European call at time t is a function of the *market state*: calendar time t and spot price S_t ; and *contract features*: maturity T and strike K

$$C_t = C(S_t, t; K; T)$$

Markov assumption?

- Call prices may (and, in practice, do) also depend on past history, as in:

$$C_t = C[S_t, t; K; T; \{S_s, s \in [0, t]\}]$$

- In this case, the call price is a *functional* of the path $\{S_s, s \in [0, t]\}$
- Classic stochastic calculus does not cover calculus of functionals: things like $\partial f(\{X_s, s \in [0, t]\})/\partial X_t$ have no meaning in Ito's framework
- The calculus of stochastic functionals, or "Ito Functional Calculus", was only discovered in 2008 by Bruno Dupire, and is (way) out of scope of this course
- One of its results is that all that follows, "as if" the call price were a function of the current spot alone, $C_t = C(S_t, t)$, also holds when it is a functional of the path – intuitively, past history is written in stone, immutable and out of our control, hence, it does not affect the arguments that follow

European Call: Pricing Function

Hand-waving argument

It is reasonable to assume that call prices mainly depend on *distance to strike* and *time to maturity*:

$$C_t = C(S_t, t; K; T) \approx g(S_t - K, T - t)$$

This is inexact, even in the context of simplest models (such as Black and Scholes), but "approximately" true and not an unreasonable approximation

Thankfully, we don't need this approximation in our derivations,
only to build intuition

European Call: Pricing Function

Properties of the pricing function

For fixed maturity T and strike K , denote the call price $C_t = C(S_t, t)$

- Price is above intrinsic: $C(S_t, t) \geq (S_t - K)^+ \geq 0$
- Asymptotic behavior: $C \rightarrow S_t - K$ when $S_t \rightarrow \infty$ and $C \rightarrow 0$ when $S_t \rightarrow -\infty$ (or 0 for positive asset prices such as equities)
- Delta $\Delta = \frac{\partial C}{\partial S_t} \geq 0$ since $\frac{\partial C}{\partial K} \leq 0$ and we assumed $C \approx g(S - k)$
- Gamma $\Gamma = \frac{\partial^2 C}{\partial S_t^2} \geq 0$ since $\frac{\partial^2 C}{\partial K^2} \geq 0$ and we assumed $C \approx g(S - k)$
- Theta $\theta = \frac{\partial C}{\partial t} \leq 0$ since $\frac{\partial C}{\partial T} \geq 0$ and we assumed $C \approx g(T - t)$

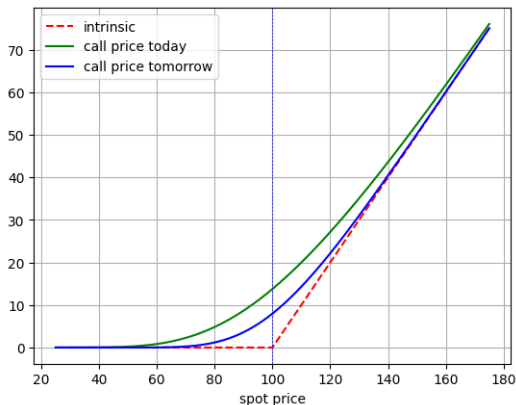
Notations

- We denote partial derivatives of functions either $\partial f / \partial x$ or f_x , and $\partial^2 f / \partial x^2$ or f_{xx} for higher-order derivatives
- For the partial derivatives of call prices, we also use market conventions: C_s is called "Delta", C_{ss} is called "Gamma", C_t is called "Theta" – those sensitivities are collectively called "Greeks"

European Call: Pricing Function

The shape of the pricing function is highly constrained by arbitrage: the prescribed asymptotes, monotonicity in spot and time, and convexity in spot, don't leave much freedom

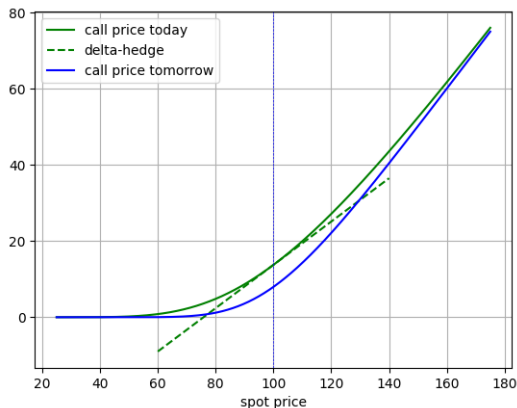
Graphically, it must be of the following shape ($S_0 = 100$, ATM $K = 100$):



Option Hedging and PnL

Delta-Hedging

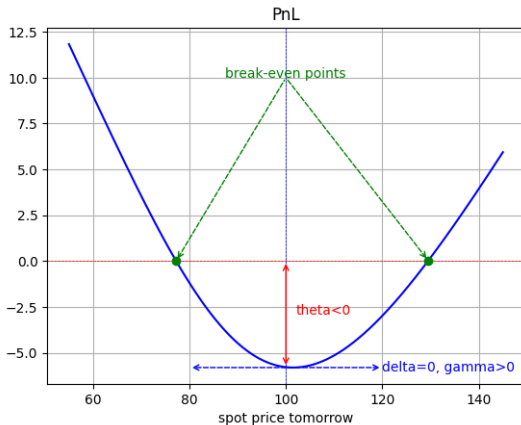
Delta-hedging is performed by selling an amount of underlying asset equal to $\Delta = \partial C / \partial S$, thereby canceling the sensitivity of the option price wrt small moves:



PnL after Hedge: Graphical Intuition

Reasoning graphically for now to build intuition:

PnL is obtained by subtracting today's value and delta-hedge (the dashed green line) from tomorrow's call price (the blue curve):



PnL after Hedge: Analytic Derivation

Time step Δt

Variation of spot price ΔS

Variation of call price by second-order Taylor expansion:

$$\Delta C = \theta \cdot \Delta t + \Delta \cdot \Delta S + \frac{1}{2} \cdot \Gamma (\Delta S)^2$$

PnL of Delta-Hedge $-\Delta \cdot \Delta S$

Combined PnL $Pnl = \theta \cdot \Delta t + \frac{1}{2} \cdot \Gamma \cdot (\Delta S)^2$

The curve of $PnL = f(\Delta S)$ is, as hinted in the graphical intuition, a positive parabola with vertex $(0, decay = \theta \cdot \Delta t < 0)$ and roots (i.e. break-even points):

$$Pnl = 0 \iff \Delta S = \pm \sqrt{\frac{-2 \cdot \theta \cdot \Delta t}{\Gamma}}$$

PnL after Hedge: Analytic Derivation

By second-order Taylor expansion in ΔS : $Pnl = \theta \cdot \Delta t + \frac{1}{2} \cdot \Gamma \cdot (\Delta S)^2$

Digression: Why a second-order expansion?

- If we are on a journey to rediscover option pricing from first principles, we really should not assume that a second-order expansion is correct – why not first-order or third-order?
- We will argue that second-order expansion is indeed correct in the following sense:
 - ▶ We can decide to delta-hedge as frequently as we want, arbitrarily shrinking the time step Δt
 - ▶ We will show that, in the limit when Δt shrinks to 0, the PnL converges to the second-order expansion, in other words, terms of order $k > 2$ in the expansion vanish, but the second-order term does not
- It should be evident that all this assumes continuous and nondeterministic price series (**why?**)
- We will also see that, under suitable regularity conditions, the semi-martingale representation is, in some sense, an inevitable consequence of arbitrage

Digression: Inevitability of semi-martingales

Argument overview

- Assuming continuous and nondeterministic price series, and mild regularity condition $\forall T \geq 0, E[S_T^2] < \infty$, we will demonstrate, by arbitrage arguments, that the *quadratic variation* $QV(0, T) = \int_0^T (dS_t)^2$ must be positive, increasing and finite:

$$\forall T_2 > T_1 \geq 0, 0 < QV(0, T_1) < QV(0, T_2) < \infty$$

- It follows, under mild regularity conditions, that quadratic variation must be a finite-variation process of the form:

$$QV(0, T) = \int_0^T \sigma_t^2 dt$$

for some positive, adapted process (σ_t^2) (which may be very general)

Digression: Inevitability of Semi-Martingales

Argument overview

- Hence, $(dS)^2/dt = \lim_{\Delta t \rightarrow 0} (\Delta S)^2/dt = \sigma_t^2$, and it follows that:
 - ▶ $\forall k > 2, \lim_{\Delta t \rightarrow 0} (\Delta S)^k/dt = \sigma_t^2 \lim_{\Delta t \rightarrow 0} (\Delta S)^{k-2} = 0$ by continuity of (S_t)
 - ▶ $\lim_{\Delta t \rightarrow 0} |\Delta S|/dt = \infty$: absolute variation is infinite or quadratic variation would be 0 by the argument above
 - ▶ Note that infinite variation is a trivial arbitrage requirement: **why?**
- Proving the correctness of the second-order expansion: as the time steps shrinks to 0, all terms of order $k > 2$ vanish but not the second-order term
- It also follows, modulo suitable regularity conditions, that (S_t) *must* be a semi-martingale of the form:

$$dS_t = \mu_t dt + \sigma_t dW_t$$

for some adapted processes (μ_t) and (σ_t) and some Brownian motion (W_t) – the semi-martingale representation, and in particular, the Brownian factor, are therefore not an assumption, but a consequence of arbitrage arguments

Digression: Inevitability of Semi-Martingales

Example: Fractional Brownian Motions

- Fractional Brownian Motions (FBM) are generalizations of Brownian Motions (BM), which attracted considerable attention in the past decade, particularly with the development of "rough volatility models" by Gatheral (2014)
- Like a BM, a FBM is a continuous, Gaussian process with zero expectation
- Unlike a BM, the increments of a FBM are not necessarily independent – the covariance function is of the shape:

$$\forall u \geq t \geq 0, E[FB_t FB_u] = (|t|^{2H} + |u|^{2H} - |u - t|^{2H})/2$$

where H is called the Hurst coefficient

- ▶ If $H = 1/2$, the FBM is a Brownian Motion
- ▶ If $H > 1/2$, increments are auto-correlated, resulting in smoother paths
- ▶ If $H < 1/2$, increments are anti-correlated, resulting in *rougher* paths
- ▶ When $H \neq 1/2$, the FBM is *not* a semi-martingale
- In particular, QV is 0 when $H > 1/2$ and infinite when $H < 1/2$
- Therefore, the only admissible model of asset prices in the FBM family is with $H = 1/2$, i.e. the Brownian Motion

Replication of Quadratic Variation

- We now prove that quadratic variation must be positive and finite (in which case, it is also, trivially, increasing)
- Consider an arbitrary partition of the time interval $[0, T]$:
 $0 = t_0 < t_1 < \dots < t_i < \dots < t_{n-1} < t_n = T$
- Denote S_i the asset price at time t_i
- Implement the classic *mean-reversion* strategy: at time t_i , buy or sell the asset so as to hold an amount proportional to the distance to $S_0 - S_i$
- This means buy when prices are low, sell when prices are high: for instance, if $S_0 = 100$ and $S_1 = 110$, sell 10 units at t_1 ; if $S_2 = 105$, record a PnL of 50 at t_2 and buy back 5 units to stay only 5 units short; if, on the contrary, the asset price decreases so that $S_1 = 97$, buy 3 units at t_1 and hold until t_2
- This strategy is profitable when asset prices mean-revert and tend to decrease from high prices or increase from low prices and remain in a range; it loses when asset prices trend and tend to increase when high or decrease when low
- The PnL from t_i to t_{i+1} is $dPnL = (S_0 - S_i)(S_{i+1} - S_i)$, hence, cumulated is $PnL = \sum_{i=0}^{n-1} (S_0 - S_i)(S_{i+1} - S_i)$

Replication of Quadratic Variation

- Therefore, we can produce an amount $\sum_{i=0}^{n-1} (S_0 - S_i)(S_{i+1} - S_i)$ with a zero-cost trading strategy
- By arbitrage, this amount must be finite, and it cannot be positive or negative with probability 1
- By arithmetic identity:

$$2 \sum_{i=0}^{n-1} (S_0 - S_i)(S_{i+1} - S_i) = \sum_{i=0}^{n-1} (S_{i+1} - S_i)^2 - (S_n - S_0)^2$$

- The term $(S_n - S_0)^2$ (called the payoff of a parabola) is positive with probability 1 (assuming continuous and nondeterministic price process) and finite with probability 1 (by regularity assumption)
- Hence, the term $\sum_{i=0}^{n-1} (S_{i+1} - S_i)^2$ must also be finite with probability 1 and positive with positive probability
- This is true for any partition of the interval $[0, T]$, so, it must be true in the limit of small time step when $\max\{t_{i+1} - t_i, 0 \leq i < n\} \rightarrow 0$ and $\sum_{i=0}^{n-1} (S_{i+1} - S_i)^2 \rightarrow \int_0^T (dSt)^2 = QV(0, T)$

Back to PnL

Summary

- We established that:

- ▶ The correct computation of delta-hedged PnL is by second-order expansion:

$$Pnl = \theta_t \cdot \Delta t + \frac{1}{2} \cdot \Gamma_t \cdot (\Delta S_t)^2$$

- ▶ It exists a finite, positive variable σ_t^2 such that:

$$(\Delta S_t)^2 = \sigma_t^2 \cdot \Delta t$$

σ_t^2 is the price variance per time unit, σ_t is called *volatility at t*

- Hence:

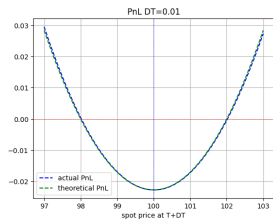
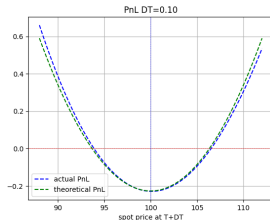
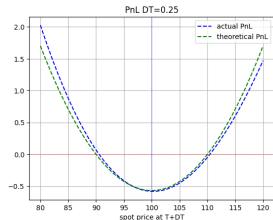
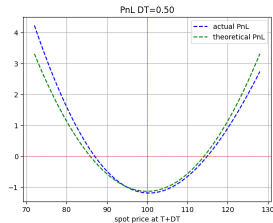
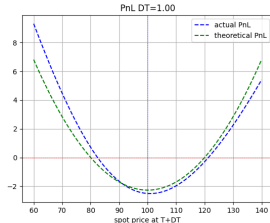
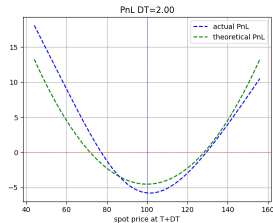
$$Pnl = (\theta_t + \frac{1}{2} \cdot \Gamma_t \cdot \sigma_t^2) \cdot \Delta t$$

and differential PnL is $dPnL/dt = \theta_t + \Gamma_t \cdot \sigma_t^2 / 2$

The above is correct in the limit of a small time mesh Δt , with interpretation: "we can re-hedge frequently enough such that PnL converges to those expressions"

European Call: PnL and break-even

The previous graphs were made with $\Delta t = 2$ years in the interest of readability – as Δt gets smaller, the actual PnL converges to the theoretical one



Black and Scholes PDE

Relationship between Volatility and Greeks

The PnL equation establishes a relationship between the volatility σ_t and the partial derivatives θ_t and Γ_t , or *Greeks*, of the pricing function

- Given volatility σ_t it must be that:

$$\frac{dPnL}{dt} = \theta_t + \frac{1}{2} \cdot \Gamma_t \cdot \sigma_t^2 = 0 \iff \Gamma_t \cdot \sigma_t^2 = -2 \cdot \theta_t$$

- ▶ If $\Gamma_t \cdot \sigma_t^2 > -2 \cdot \theta_t$ then high volatility moves the asset price outside of break-even, making a positive profit for the holder
- ▶ If $\Gamma_t \cdot \sigma_t^2 < -2 \cdot \theta_t$ then low volatility traps the asset price within break-even, making a positive profit for the writer
- ▶ A pricing function consistent with given volatility σ_t must be such that, given volatility, the asset price lands *on* a break-even point, i.e:

$$\Gamma_t \cdot \sigma_t^2 = -2 \cdot \theta_t$$

- Conversely, a given pricing function $C(S_t, t)$ *implies* volatility: $\hat{\sigma}_t = \sqrt{\frac{-2C_t}{C_{ss}}}$, the volatility needed to cancel PnL after hedge, when PnL is computed with this pricing function

Black and Scholes PDE

This relationship we just established between volatility and Greeks:

$$\frac{1}{2}\Gamma\sigma^2 = -\theta$$

was first derived by Black and Scholes in 1973

- The LHS is the dollar-value of convexity – delta-hedge monetizes convexity, measured by Γ , because the delta-hedge of a convex profile involves buying low and selling high, thereby capturing the oscillations of the asset price
- Convexity value is directly proportional to the variance σ^2 , which measures the magnitude of the price oscillations
- The equation states that the decay on the RHS, which measures time value over a small time step Δt , must be equal to the value of convexity during the time step
- This equation turns out to be very general and establishes a sort of "universal law of nonlinear financial instruments": time value equals half convexity times variance – a triumph of Black and Scholes ground-breaking paper

Black and Scholes PDE

$$\frac{1}{2}\Gamma\sigma^2 = -\theta$$

- This is a Partial Differential Equation (PDE), stating a relationship between some partial derivatives of the pricing function $C(S, t)$, which must be satisfied at all times to prevent arbitrage *given volatility*
- The existence and uniqueness of solutions to such PDEs are well-known, given appropriate *boundary conditions* – for the European call, the obvious condition is that price coincides with payoff at maturity:

$$C(S, T) = (S - K)^+$$

- Importantly, this equation is not really a "pricing" equation, all it does is establish a one-to-one relationship between the volatility model (σ_t) and the (partial derivatives of the) pricing function $C(S, t)$
- The solution can be found analytically, numerically, or approximately, depending on the volatility model

Volatility Models

Volatility Models

- We established a one-to-one mapping, in the form of a PDE, between volatility and the (derivatives of) the pricing function
- We did so in a *model-free* manner, without assuming anything specific regarding the probabilistic dynamics of asset prices: we showed that (modulo regularity conditions) it must exist an adapted volatility process (σ_t), but we didn't assume anything about it
- This is as far as we can go without saying something specific about volatility
- This is performed with a *model*, a mathematical simplification of a complex reality, designed to facilitate the understanding and control of nontrivial phenomena, and the solution of difficult problems
- Model selection and design is as much an art as a science: we want to disregard irrelevant details and highlight what matters most for a given problem (such as: pricing, market making, risk management, risk taking...) and facilitate the discovery of effective solutions
- the "right" model very much depends on goals and context – in Einstein's words "make it as simple as possible, but not simpler"

Volatility Models

Established volatility models

Bachelier $\sigma_t = \sigma_N$ a known constant

Black and Scholes $\sigma_t = \sigma_L \cdot S_t$ where σ_L is a known constant

Time-Dependent variants where Bachelier's σ_N or Black and Scholes' σ_L are deterministic functions of calendar time t

CEV (Constant Elasticity Volatility) $\sigma_t = \sigma_\beta S_t^\beta$
where σ_β (and β) are constants or functions of time

Affine Volatility (also called Displaced-Lognormal) $\sigma_t = a + bS_t$
where a and b are constants or functions of time
notice that CEV and Displaced families of models both include Bachelier and Black and Scholes

Homogeneous Diffusion $\sigma_t = \sigma_{LV}(S_t)$ deterministic function of spot,
generalizes previous models

General Diffusion (also called Dupire model or local volatility model)
 $\sigma_t = \sigma_{LV}(S_t, t)$ deterministic function of spot and time

Volatility Models

More sophisticated volatility models

- In all these models, the asset price (S_t) is a self-contained diffusion (modulo the drift term, which, as we will see, does not matter)
- Dupire's model is, by definition, the most general diffusion model
- We only cover diffusion models, but asset prices don't have to be diffusions, they only have to be semi-martingales (when continuous)
- Popular models where asset prices are *not* self-contained diffusions include:
 - ▶ Stochastic Volatility Models such as SABR (Hagan, 1992) or Heston (1993), where volatility follows its own diffusion, driven by another (possibly correlated) Brownian motion
 - ▶ More realistic *Path-Dependent* Volatility models (Gyon, 2014) where volatility is a function(al) of the path: $\sigma_t = g(\{S_s, 0 \leq s \leq t\})$
 - ▶ Stochastic-Local-Volatility (SLV) that combine stochastic and local volatility: $\sigma_t = \sigma_{LV}(S_t, t)\xi_t$, where (ξ_t) is another diffusion
 - ▶ Rough Volatility Models (Gatheral, 2014), where volatility is not a diffusion, or even a semi-martingale, but a more general process driven by a *Fractional* Brownian Motion

Normal and Lognormal Volatility

- We defined volatility σ_t by: $(dS_t)^2 = \sigma_t^2 dt$
- This is the *normal* volatility, the (annualized) standard deviation of *first differences* dS_t – sometimes denoted σ_t^N
- Some literature works with the *lognormal* volatility, defined by $(dS_t/S_t)^2 = (d \log S_t)^2 = \sigma_t^L dt$, the standard deviation of *returns*, denoted σ_t^L
- Lognormal volatility is only well-defined for positive price processes (such as equities or foreign exchange rates), not for those that may cross zero (such as interest rates or commodity futures) – conversely, normal volatility is always well-defined, hence, more general – for positive processes: $\sigma_t^N = \sigma_t^L S_t$
- For positive processes, working with normal or lognormal volatility is a matter of convention: the same process can be written both ways, either $dS_t = \dots dt + \sigma_t^N dW_t$ or $dS_t/S_t = \dots dt + \sigma_t^L dW_t$ where $\sigma_t^L = \sigma_t^N/S_t$
- Do not confuse convention (how we write the model) with modeling (what the model assumes): Bachelier's model assumes that normal volatility is constant, Black and Scholes assumes that lognormal volatility is constant, meaning normal volatility is proportional – these are not different conventions but very different models that produce different option prices

Risk-Neutral Pricing

Solving the pricing PDE

Pricing PDE

- For a fixed strike K and maturity T , we went on a journey to derive the pricing function $C(S_t, t; K, T)$, simply denoted $C(S_t, t)$, for the European call at time t function of the spot price S_t
- We derived the PDE: $C_{ss}(S_t, t)\sigma_t^2 = -2C_t(S_t, t)$
- We restricted the scope to self-contained diffusions, where $\sigma_t^2 = \sigma(S_t, t)$, resulting in the PDE:

$$C_{ss}(S_t, t)\sigma^2(S_t, t) = -2C_t(S_t, t)$$

with boundary $C(S_T, T) = (S_T - K)^+$

- This PDE can be solved numerically with finite-difference methods (FDM)
- Under simple volatility specifications, such as Bachelier $\sigma(S_t, t) = \sigma$ or Black-Scholes $\sigma(S_t, t) = \sigma S_t$ it can be solved analytically with techniques generally taught in Physics class
- The Feynman-Kac theorem provides the means of solving these PDEs with probabilistic tools

Feynman-Kac Theorem

The Feynman-Kac theorem states that solutions of PDEs must be expectations, so that PDEs are solved by computing expectations!

Theorem (Feynman-Kac)

The function of two real variables $f(x, y)$ satisfies the PDE:

$$f_{xx}(x, y)h(x, y)^2 = -2f_y$$

under the boundary condition:

$$f(x, Y) = g(x)$$

where $g(x)$ and $h(x, y)$ are arbitrary functions

if and only if

$$f(x, y) = E[g(X_Y)|X_y = x]$$

where the expectation is taken under the probabilistic dynamics:

$$dX_t = h(X_t, t)dW_t$$

and (W_t) is a standard Brownian motion

Feynman-Kac Theorem

Proof.

- Assuming $dX_t = h(X_t, t)dW_t$, we check that $f(x, t) = E[g(X_T)|X_t = x]$ satisfies the PDE and boundary – standard PDE analysis shows that the solution must be unique, concluding the proof
- Boundary is trivial: $f(x, T) = E[g(X_T)|X_T = x] = g(x)$
- By Ito's lemma: $df(X_t, t) = f_x(X_t, t)dX_t + [f_t(X_t, t) + \frac{1}{2}f_{xx}(X_t, t)h(X_t, t)]dt$
Hence: $E_t[df(X_t, t)] = f_x(X_t, t)E_t[dX_t] + [f_t(X_t, t) + \frac{1}{2}f_{xx}(X_t, t)h(X_t, t)]dt$
- Notice that (X_t) is a driftless diffusion, hence, a martingale, so:

$$E_t[dX_t] = 0 \text{ hence } f_t(X_t, t) + f_{xx}(X_t, t)h(X_t, t)/2 = E_t[df(X_t, t)]/dt$$

- Denote $M_t = f(X_t, t)$ – we show that (M_t) is a martingale, so $E_t[dM_t] = 0$, hence, $f_t(X_t, t) + f_{xx}(X_t, t)h(X_t, t)/2 = 0$, proving that f satisfies the PDE
- $\forall u \geq t \geq 0$:

$$E_t[M_u] = E_t[f(X_u, u)] = E_t[E_u[g(X_T)]] = E_t[g(X_T)] = f(X_t, t) = M_t$$

where we used the Markov property of the diffusion (X_t) and the boxed expectation formula – hence, (M_t) is a martingale



Consequences of Feynman-Kac

Pricing by expectation

With the restriction $\sigma_t = \sigma(S_t, t)$, we proved that:

$$C(S_t, t; K, T) = E[(S_T - K)^+ | S_t]$$

where the expectation is computed under the probabilistic dynamics:

$$dS_t = \sigma(S_t, t)dW_t$$

Notice that, under this dynamics, both the underlying price S_t and the call price $C_t = C(S_t, t)$ are both martingales

This result holds (in some sense to be precisely defined) for more general volatility models, although we don't prove this statement

Consequences of Feynman-Kac

Risk-neutral pricing

- The pricing dynamics $dS_t = \sigma(S_t, t)dW_t$ has nothing to do with the real world, it only exists as a PDE solving trick
- Recall that the price series (S_t) must be a semi-martingale, hence, assuming $\sigma_t = \sigma(S_t, t)$ is the same as assuming that the *real-world* process is:

$$dS_t = \mu_t dt + \sigma(S_t, t)dW_t$$

for some drift process (μ_t)

- Hence, the "pricing" dynamics, has the same volatility than real-world dynamics, but without drift
- This is a consequence of delta-hedging, which cancels exposure to the drift, but not to volatility, due to the non-linearity of call prices
- It follows that the real-world drift (μ_t) is irrelevant for option pricing, only volatility matters – this is a nontrivial result with massive consequences, and a triumph of Black and Scholes' 1973 ground-breaking work

Consequences of Feynman-Kac

Risk-neutral pricing

- In Economics, the drift of an asset price is also called "market price of risk" and interpreted as a reward for investing in risky assets
- For this reason, the "pricing" dynamics, where the drift is zeroed, is called *risk-neutral SDE* and the corresponding probability measure (which emphatically does not reflect real-world probabilities) is called *risk-neutral measure*
- Under the risk-neutral probability measure, prices coincide with expectations, both for underlying assets and for Derivatives contracts
- The risk-neutral measure is conventionally denoted Q by opposition to the physical measure P

Constant Volatility Models

Bachelier and Black-Scholes Models

Bachelier's Model

Assumption $\sigma_t = \sigma$ constant normal volatility

Support $S_t \in (-\infty, \infty)$

Risk-neutral SDE $dS_t = \sigma dW_t^Q$

Solution $S_T = S_t + \sigma(W_T - W_t) \sim N(S_t, \sigma\sqrt{T-t})$

Call price $C(S_t, t; K, T) = E_t^Q[(S_T - K)^+] =$
 $\int_{\mathbb{R}} (x - K)^+ n(x; S_t, \sigma\sqrt{T-t}) dx$

Formula $C(S_t, t; K, T) = \text{Bach}(S_t, t; K, T; \hat{\sigma}) =$
 $(S_t - K)N(d) + \hat{\sigma}\sqrt{T-t}n(d)$
where $d = \frac{S_t - K}{\hat{\sigma}\sqrt{T-t}}$

Bachelier and Black-Scholes Models

Black and Scholes' Model

Assumption $\sigma_t = \sigma S_t$ constant lognormal / proportional normal volatility

Support $S_t \in (0, \infty)$

Risk-neutral SDE $\frac{dS_t}{S_t} = \sigma dW_t^Q$

Solution $\log S_T = \log S_t - \frac{\sigma^2}{2}(T-t) + \sigma(W_T - W_t) \sim N[\log S_t - \frac{\sigma^2}{2}(T-t), \sigma\sqrt{T-t}]$

Call price $C(S_t, t; K, T) = E_t^Q[(S_T - K)^+] = \int_{\mathbb{R}} (e^x - K)^+ n[x; \log S_t - \frac{\sigma^2}{2}(T-t), \sigma\sqrt{T-t}] dx$

Formula $C(S_t, t; K, T) = BS(S_t, t; K, T; \hat{\sigma}) = S_t N(d_1) - KN(d_2)$
where $d_{1/2} = \frac{\log \frac{S_t}{K} \pm \frac{\hat{\sigma}^2}{2}(T-t)}{\hat{\sigma}\sqrt{T-t}}$

Bachelier and Black-Scholes Models

Greeks

	Bachelier	Black-Scholes
$C(S_t, t; K, T; \hat{\sigma})$	$(S_t - K)N(d) + \hat{\sigma}\sqrt{\tau}n(d)$	$S_tN(d_1) - KN(d_2)$
$\Delta = C_s = \partial C / \partial S_t$	$N(d)$	$N(d_1)$
$\Gamma = C_{ss} = \partial^2 C / \partial S_t^2$	$n(d) / \sigma\sqrt{\tau}$	$n(d_1) / S_t\sigma\sqrt{\tau}$
$\theta = C_t = \partial C / \partial t$	$-\sigma n(d) / 2\sqrt{\tau}$	$-S_t\sigma n(d_1) / 2\sqrt{\tau}$
$\text{vega} = C_\sigma = \partial C / \partial \hat{\sigma}$	$n(d)\sqrt{T - t}$	$n(d_1)S_t\sqrt{\tau}$

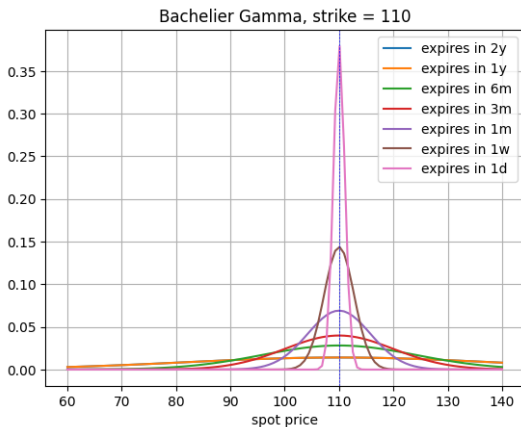
$$\tau = T - t, d = \frac{S_t - K}{\hat{\sigma}\sqrt{\tau}}, d_{1/2} = \frac{\log \frac{S_t}{K} \pm \frac{\hat{\sigma}^2}{2}(\tau)}{\hat{\sigma}\sqrt{\tau}}$$

Gamma

- Recall, gamma measures convexity, with is how options monetize volatility
- In Bachelier: $\Gamma = n(S_t; K, \sigma\sqrt{T-t})$
this is the probability density of a scaled Brownian motion (S_t) with volatility σ , moving *backward in time* from boundary $S_T = K$
- It follows that gamma is maximum around the money, and close to expiry
- Hence, long European call positions capture volatility in a non-uniform manner: their exposure to realized volatility is maximized when the spot price S_t hovers around the strike K , particularly near expiry when $t \rightarrow T$
- We will soon formalize this important point when we derive the mis-replication formula
- A similar analysis holds for Black and Scholes, modulo the log convexity bias

Gamma

$$\Gamma = n(S_t; K, \sigma\sqrt{T-t})$$



Implied Volatility

- Recall, our pricing machinery derives call prices as a function of volatility
- Bachelier and Black and Scholes pricing formulas are increasing functions of volatility: $\text{vega} > 0$
- Hence, there is a one-to-one mapping between volatility and call price
- This mapping can be inverted (generally, numerically) to compute the *volatility implied by a given price*
- More formally, given the price of a European call of strike K and maturity T when spot price is S_t at time t :
 - ▶ The *normal implied volatility* is the unique volatility fed to Bachelier's formula to match a given price:

$$\text{Bach}(S_t, t; K, T; \hat{\sigma}_N) = C_t$$

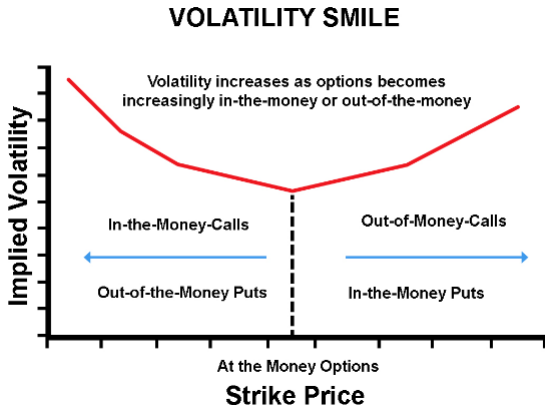
- ▶ The *lognormal implied volatility* is the unique volatility fed to Black and Scholes' formula to match a given price:

$$\text{BS}(S_t, t; K, T; \hat{\sigma}_L) = C_t$$

Market Implied Volatility Smile

- On some active option markets such as S&P options on the CBOE, options of various strikes and maturities trade transparently in large volumes, with high liquidity
- The implied (say, lognormal) volatilities of all those options are generally wildly different across strikes and maturities
- In particular, implied volatilities tend to increase away from the money, so the graph of implied volatility function of strike looks like a smiley – the name stuck
- This of course doesn't make much sense at first sight: how can different options on the same underlying asset reflect different assumptions regarding the probabilistic behavior of the asset?
- The existence of implied volatility smiles highlights the inadequacy of the simplest pricing models to correctly represent a complex reality

Market Implied Volatility Smile



source: quantra.quantinsti.com/glossary/Volatility-Smile

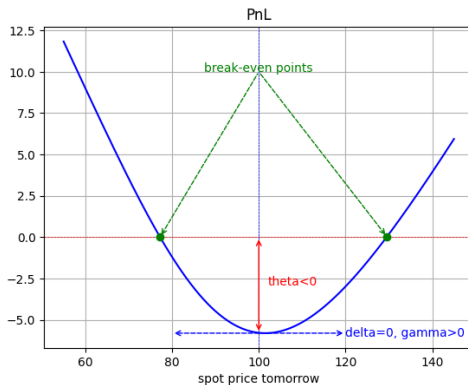
Inadequacy of simple models?

- Many authors today consider Bachelier and Black and Scholes models as glorified conversions between dollar-price and implied volatility units, and focus on more sophisticated models, such as Dupire, that make better sense of observed option prices
- Rebonato famously wrote " "implied volatility is the wrong number you put in the wrong formula to get the right price"
- Traders routinely quote and trade options in volatility rather than price units
- It is rather underwhelming that such efforts and ingenuity were deployed to produce some kind of conversion
- We will show that there is much more to Bachelier and Black and Scholes than some conversion, that implied volatility has important, physical semantics
- We follow the works of El Karoui (Robustness of Black and Scholes, 1998) Poulsen (Fundamental Theorem of Options Trading, 2015)
- Leading us towards the key mis-replication formula

Mis-Replication Formulas

Back to PnL again

We start building intuition from the PnL graph established earlier:



We now know that (modulo a small enough time step Δt) that $PnL = \theta \cdot \Delta t + \frac{1}{2} \cdot \Gamma \cdot (\Delta S)^2$ and break even points are specified by:

$$PnL = 0 \iff \Delta S = \pm \sqrt{\frac{-2 \cdot \theta \cdot \Delta t}{\Gamma}}$$

Implied Volatility and Break-Even

- Consider now that we *manage* the position with Bachelier's model and a given implied volatility $\hat{\sigma}$
- Then our price and Greeks satisfies Bachelier's PDE $-\theta = \Gamma \cdot \hat{\sigma}^2/2$ *with our implied volatility*
- Substituting θ in the PnL formula:

$$PnL = \Gamma \cdot [(\Delta S)^2 - \hat{\sigma}^2 \Delta t]/2$$

- Implied variance is what one pays to receive the *delivered* variance $(\Delta S)^2$
- Trading options is swapping implied for delivered volatility (weighted by gamma)
- In particular, the break-even points directly reflect implied volatility:

$$PnL = 0 \iff \Delta S = \pm \hat{\sigma} \sqrt{\Delta t}$$

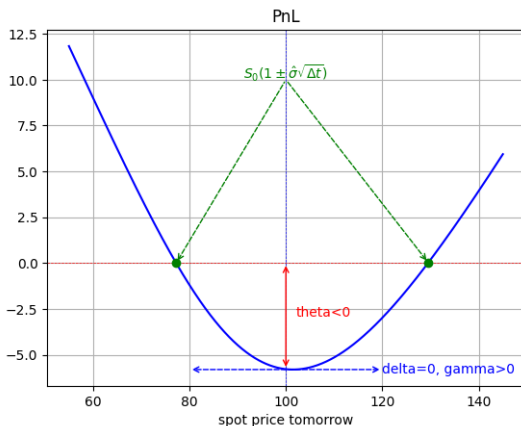
- Similarly, in Black and Scholes:

$$PnL = \Gamma \cdot S^2 \cdot [(\Delta S/S)^2 - \hat{\sigma}^2 \Delta t]/2 = 0 \iff \Delta S/S = \pm \hat{\sigma} \sqrt{\Delta t}$$

- Implied volatility is a tradable market estimate of future volatility

Implied Volatility and Break-Even in Black and Scholes

$$PnL = \Gamma \cdot S^2 \cdot [(\Delta S/S)^2 - \hat{\sigma}^2 \Delta t]/2 = 0 \iff \Delta S/S = \pm \hat{\sigma} \sqrt{\Delta t}$$



Note that break-even are not symmetric on the graph because Δt is too large

Drift does not matter

- This graph gives us another interpretation why drift does not matter
- Break-even points are proportional to $\sqrt{\Delta t}$
- Drift, by definition, is proportional to Δt
- Hence, however large the drift may be compared to volatility:
with small enough time step, its impact on PnL must be negligible compared to volatility
- Giving us another interpretation of Black and Scholes' ground-breaking result, directly related to real-world PnL: impact of the drift must vanish with frequent enough hedging

(Mis-)Replication Formula

- Recall the real-world price series (S_t) must be a semi-martingale of the form:

$$dS_t = \cdots dt + \sigma_t^r dW_t^P$$

where the "r" subscript emphasizes that this is the delivered, real-world volatility and the "P" subscript denotes the physical measure

- We manage a European call with Bachelier and a given implied volatility $\hat{\sigma}$ – applying Ito's lemma to our pricing function, under the real-world dynamics:

$$dC_B(S_t, t) = \Delta_B(S_t, t)dS_t + [\theta_B(S_t, t) + \frac{1}{2}\Gamma_B(S_t, t)(\sigma_t^r)^2]dt$$

where the subscript "B" denotes Bachelier

- But of course: $\theta_B(S_t, t) = -\Gamma_B(S_t, t)\hat{\sigma}^2/2$ – substituting:

$$dC_B(S_t, t) = \Delta_B(S_t, t)dS_t + \frac{1}{2}\Gamma_B(S_t, t)[(\sigma_t^r)^2 - \hat{\sigma}^2]dt$$

- Integrating from 0 to T and noting that $C_B(S_T, T) = (S_T - K)^+$, we find:

$$(S_T - K)^+ = C_B(S_0, 0) + \int_0^T \Delta_B(S_t, t)dS_t + \frac{1}{2} \int_0^T \Gamma_B(S_t, t)[(\sigma_t^r)^2 - \hat{\sigma}^2]dt$$

(Mis-)Replication Formulas

Generalization

- The formula easily generalizes to arbitrary models (M) of the form:

$$dS_t = \sigma_M(S_t, t) dW_t:$$

$$(S_T - K)^+ =$$

$$C_M(S_0, 0) + \int_0^T \Delta_M(S_t, t) dS_t + \frac{1}{2} \int_0^T \Gamma_M(S_t, t) [(\sigma_t^r)^2 - \sigma_M^2(S_t, t)] dt$$

- For instance, in Black and Scholes:

$$(S_T - K)^+ = C_{BS}(S_0, 0) + \int_0^T \Delta_{BS}(S_t, t) dS_t + \frac{1}{2} \int_0^T \Gamma_{BS}(S_t, t) S_t^2 [(\sigma_L^r)^2 - \hat{\sigma}_L^2] dt$$

- Without the rightmost integral, this is the replication formula found in textbooks:
delta-hedge replicates the payoff given initial funds equal to today's price
- But of course this only works when volatility is correctly anticipated
in which case the rightmost integral is 0
- The rightmost integral is the *mis-replication* term traders experience every day: it is impossible to correctly anticipate delivered volatility

Mis-Replication Formula

General:

$$(S_T - K)^+ = C_M(S_0, 0) + \int_0^T \Delta_M(S_t, t) dS_t + \frac{1}{2} \int_0^T \Gamma_M(S_t, t) [(\sigma'_t)^2 - \sigma_M^2(S_t, t)] dt$$

With M = Bachelier:

$$(S_T - K)^+ = C_B(S_0, 0) + \int_0^T \Delta_B(S_t, t) dS_t + \frac{1}{2} \int_0^T \Gamma_B(S_t, t) [(\sigma'_t)^2 - \hat{\sigma}^2] dt$$

With M = Black-Scholes:

$$(S_T - K)^+ = C_{BS}(S_0, 0) + \int_0^T \Delta_{BS}(S_t, t) dS_t + \frac{1}{2} \int_0^T \Gamma_{BS}(S_t, t) S_t^2 [(\sigma'_L)^2 - \hat{\sigma}_L^2] dt$$

Mis-replication Term

$$\frac{1}{2} \int_0^T \Gamma_M(S_t, t) [(\sigma_t^r)^2 - \sigma_M^2(S_t, t)] dt$$

With M = Bachelier: $\frac{1}{2} \int_0^T \Gamma_B(S_t, t) [(\sigma_t^r)^2 - \hat{\sigma}^2] dt$

With M = Black-Scholes: $\frac{1}{2} \int_0^T \Gamma_{BS}(S_t, t) S_t^2 [(\sigma_t^r)^2 - \hat{\sigma}_L^2] dt$

- Accumulates differences between delivered and implied variance
- Gamma-weighted
- Hence, what affects PnL most is realized volatility *when spot market hovers around the money*, especially near maturity
- Finite-variation process: positions managed with mis-calibrated models do not "explode", they slowly bleed to death

Local Volatility Models

Time-dependent volatility

Bachelier: $dS_t = \sigma(t)dW_t^Q$, Black-Scholes: $dS_t/S_t = \sigma_L(t)dW_t^Q$

- Recall from Stochastic Calculus class that if $dX_t = \sigma(t)dW_t$ then:

$$X_T \sim N[X_0, \int_0^T \sigma(t)^2 dt]$$

- This is the same distribution, hence, same call prices *for maturity T only* as if:

$$dX_t = \hat{\sigma}(T)dW_t \text{ with } \hat{\sigma}(T) = \sqrt{\frac{\int_0^T \sigma^2(t)dt}{T}}$$

- Hence, pricing in time-dependent models simply reuse formulas of constant volatility models, with implied volatility = quadratic average local volatility:

$$\hat{\sigma}^2(T) = \frac{\int_0^T \sigma^2(t)dt}{T} \iff \sigma^2(t) = \frac{\partial[t\hat{\sigma}^2(t)]}{\partial t} = \hat{\sigma}^2(t) + t\hat{\sigma}'^2(t)$$

- Used both ways: to compute prices/implied volatilities given anticipated volatility $\sigma(t)$ over time,
or *calibrate* local volatility $\sigma(t)$ to a market implied volatility curve $\hat{\sigma}(T)$

Spot-dependent volatility – Dupire's model

$$dS_t = \sigma(S_t, t) dW_t^Q$$

- In time-dependent models, implied variance is the average local variance across time
- Can we say something similar with spot-dependent models?
- The answer is yes: implied variance is still an average local variance, across time *and asset prices*, *weighted by probability and gamma*
- Normal implied volatility formula:

$$\hat{\sigma}_N^2(K, T) = \frac{E^{Q(LV)}\{\int_0^T \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)] \sigma^2(S_t, t) dt\}}{E^{Q(LV)}\{\int_0^T \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)] dt\}}$$

where $Q(LV)$ is the risk-neutral measure *under the local volatility model* and Γ_B is gamma, computed in Bachelier

- And (very) similar for the lognormal implied

Lemma

$$\hat{\sigma}_N^2(K, T) = \frac{E^{Q(LV)}\{\int_0^T \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)]\sigma^2(S_t, t)dt\}}{E^{Q(LV)}\{\int_0^T \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)]dt\}}$$

Lemma

- Consider two different models:

$$M_1 : dS_t = \sigma_1(S_t, t)dW_t^{Q_1}, M_2 : dS_t = \sigma_2(S_t, t)dW_t^{Q_2}$$

where Q_i is the risk-neutral measure under model i

- Denote $C_i(S_t, t; K, T)$ the call pricing function in model i , then the price difference $C_2(S_t, t; K, T) - C_1(S_t, t; K, T)$ is:

$$E^{Q_2}\left\{\int_t^T \frac{\Gamma_1(S_u, u; K, T)}{2} [\sigma_2^2(S_u, u) - \sigma_1^2(S_u, u)] du\right\}$$

- Proof of lemma is a direct consequence of the mis-replication formula
- Proof of the main formula and lognormal variant:
set $M_1 =$ Bachelier or Black and Scholes, and $M_2 =$ Dupire

Gamma-Weighted Average Formula

$$\hat{\sigma}_N^2(K, T) = \frac{E^{Q(LV)}\{\int_0^T \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)]\sigma^2(S_t, t)dt\}}{E^{Q(LV)}\{\int_0^T \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)]dt\}}$$

- Denote the weights $w(S_t, t)$ of local variance in the implied variance formula:

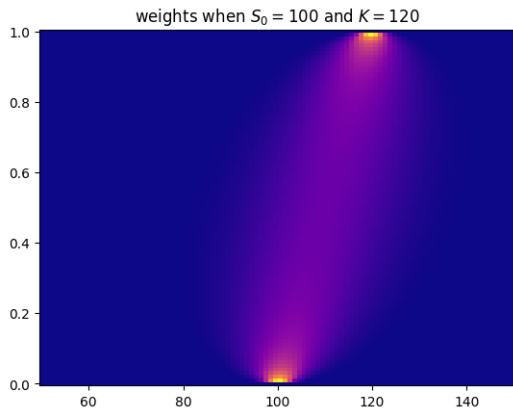
$$w(S_t, t) = \phi_{LV}(S_t, t) \cdot \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)]$$

where $\phi_{LV}(x, t)$ is the risk-neutral density of $S_t = x$ in the LV model

- Then: $\hat{\sigma}_N^2(K, T) = \frac{\int_0^T \int_{\mathbb{R}} w(x, t) \sigma^2(x, t) dx dt}{\int_0^T \int_{\mathbb{R}} w(x, t) dx dt}$
normal implied variance is effectively a weighted average of local variances
- Recall $\Gamma_B = n(S_t; K, \sigma\sqrt{T-t})$ is the density of a scaled Brownian motion, moving backward in time from K at T , maximized on strike at maturity
- By contrast, ϕ_{LV} is the density of the diffusion, moving forward in time from S_0 today, maximized on current spot today
- Hence, weights are concentrated on the line between $(S_0, 0)$ and (K, T)

Weight Map

$$w(S_t, t) = \phi_{LV}(S_t, t) \cdot \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)]$$



Implied variance is essentially a line integral of local variances along the line that joins $(S_0, 0)$ and (K, T)

Time-Homogeneous LV Models

$$dS_t = \sigma(S_t) dW_t^Q$$

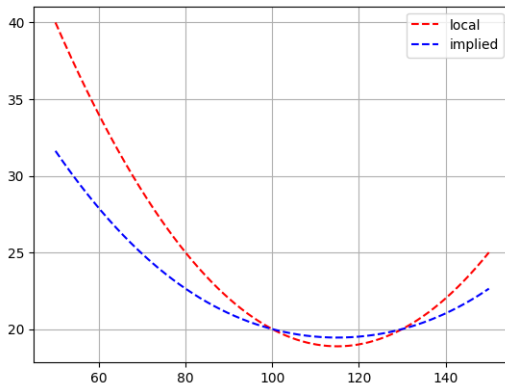
- From previous analysis, we know that $\hat{\sigma}(K, T)$ is a weighted (quadratic) average of $\sigma(S_t)$, with weights concentrated on the interval between S_0 and K , peaking at S_0 and K
- (Very) roughly: $\hat{\sigma}(K, T) \approx \sqrt{\frac{\sigma^2(S_0) + \sigma^2(K)}{2}}$
- Implied volatility damps local volatility by averaging
conversely, local volatility amplifies the oscillations of implied volatility
- In particular:

$$\hat{\sigma}(S_0, T) \approx \sigma(S_0) \text{ and } \hat{\sigma}_K(S_0, T) \approx \frac{1}{2} \sigma_s(S_0)$$

in traders parlance, the implied *skew* (ATM slope of implied) is around half the *backbone* (how the model expects ATM to react to small spot moves)

Local and Implied Volatility

$$\hat{\sigma}(K, T) \approx \sqrt{\frac{\sigma^2(S_0) + \sigma^2(K)}{2}}$$



Dupire's Formula

In a local volatility model $dS_t = \sigma(S_t, t)dW_t^Q$

- We want to establish a one-to-one relationship between:
 - ▶ Today's call prices $C(S_0, 0; K, T)$ or equivalently implied volatilities $\hat{\sigma}(K, T)$
 - ▶ And local volatilities $\sigma(S_t, t)$
- We found a formula linking the two:

$$\hat{\sigma}_N^2(K, T) = \frac{E^{Q(LV)}\{\int_0^T \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)]\sigma^2(S_t, t)dt\}}{E^{Q(LV)}\{\int_0^T \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)]dt\}}$$

- This formula is helpful for reasoning about local and implied volatilities:
 - ▶ Shows that implied volatilities are weighted (quad) averages of local volatilities
 - ▶ Relates prices to expected convexity PnL
 - ▶ Provides approximate tools to relate smile to backbone
- But not practical: implicit ($\hat{\sigma}$ appears on both sides) and hard to compute
- What is very practical (if maybe less intuitive) is Dupire's celebrated formula:

$$2C_T(S_0, 0; K, T) = \sigma^2(K, T)C_{kk}(S_0, 0; K, T)$$

Dupire's dual PDE

$$2C_T(S_0, 0; K, T) = \sigma^2(K, T)C_{kk}(S_0, 0; K, T)$$

Pricing: $\sigma \rightarrow \hat{\sigma}$

- We want to compute prices of calls $C(S_0, 0; K, T)$
or, equivalently, their implied volatilities $\hat{\sigma}(S_0, 0; K, T)$
- Given some anticipated local volatility $\sigma(S_t, t)$
- We could use Black and Scholes PDE for a fixed strike K and expiry T :

$$C_t = -C_{ss}\sigma^2/2 \text{ with final boundary } C(x, T) = (x - K)^+$$

Start at the final boundary $t = T$, and work your way backwards to today, computing the prices $C(S_t, t; K, T)$ of *one* option (K, T fixed) across calendar time and market scenarios (as a function of S_t AND t)

- Or, we can use Dupire's *dual* PDE for fixed S_0 at $t = 0$:

$$C_T = C_{kk}\sigma^2/2 \text{ with initial boundary } C(x, 0) = (S_0 - x)^+$$

Start at the *initial* boundary $T = 0$, and work your way *forward*, pricing *all* options of all strikes K and expiries T , in *one configuration* S_0 at 0

Dual PDEs

Two computations for $C(S_t, t; K, T)$ in the local volatility model

	Black and Scholes (Feynman-Kac)	Dupire (Fokker-Planck)
S_t, t domain	$\forall S_t \in \mathbb{R}_{(+)}, t \in [0, T]$	fixed $t = 0$ and S_0
K, T domain	fixed K, T	$\forall T \in \mathbb{R}_+, K \in \mathbb{R}_{(+)}$
PDE	$C_t = -C_{ss}\sigma^2/2$	$C_T = C_{kk}\sigma^2/2$
Boundary	for $t = T$: $C(S_T, T; K, T) = (S_T - k)^+$	for $T = 0$: $C(S_0, 0; K, 0) = (S_0 - K)^+$
Direction	backward	forward
Best when	pricing one option across its lifetime (risk scenarios, early exercise)	pricing multiple options in today's market conditions (book valuation, calibration)

Dupire's dual PDE

$$2C_T(S_0, 0; K, T) = \sigma^2(K, T)C_{kk}(S_0, 0; K, T)$$

Calibration: $\hat{\sigma} \rightarrow \sigma$

- We observe today's option prices of various strikes K and expiries T : $C(S_0, 0; K, T)$ or, equivalently, their implied volatilities $\hat{\sigma}(S_0, 0; K, T)$
- We want to find the local volatility parameterization: the local volatilities $\sigma(S_t, t)$ consistent with the observed prices
- This is directly given by Dupire's formula: $\sigma(K, T) = \sqrt{\frac{2C_T(S_0, 0; K, T)}{C_{kk}(S_0, 0; K, T)}}$
- This *calibrates* the model to be used for more sophisticated instruments
- And implies a local volatility *surface* $\sigma(S_t, t)$ from a price surface $C(K, T)$ in the same way simple models imply one volatility from one price
- Dupire's model is the simplest model provably consistent with the entire market implied volatility smile

Proof of Dupire's formula

- Dupire's original (and ground-breaking) 1992 paper derives the proof from the Fokker-Planck equation
- Savine (2001) derives it in a terser manner, using generalized derivatives
- We derive it in yet another manner, from our mis-replication formula, more precisely its lemma:

$$C_2 - C_1 = E^{Q_2} \left\{ \int_t^T \frac{\Gamma_1(S_u, u; K, T)}{2} [\sigma_2^2(S_u, u) - \sigma_1^2(S_u, u)] du \right\}$$

where $C_i(S_t, t; K, T)$ is the call price in model M_i : $dS_t = \sigma_i(S_t, t)dW_t$

- Thereby deriving Dupire's formula directly from PnL considerations!
- Note that this (unpublished) proof is also due to Dupire

Proof of Dupire's formula

$$C_2 - C_1 = E^{Q_2} \left\{ \int_t^T \frac{\Gamma_1(S_u, u; K, T)}{2} [\sigma_2^2(S_u, u) - \sigma_1^2(S_u, u)] du \right\}$$

- Let model 1 be the deterministic model with 0 volatility
- Then $\sigma_1(S_t, t) = 0$ and $C_1(S_t, t; K, T) = (S_t - K)^+$
hence, $\Gamma_1(S_t, t; K, T) = \delta(S_t - K)$ the Dirac delta
- Let model 2 be the local volatility model: $C_2 = C$, $\sigma_2 = \sigma$ and $Q_2 = Q$
- $C_2(S_t, t; K, T) = C(S_t, t; K, T)$ is the price in the local volatility model
- Let $t = 0$ and substitute:

$$\begin{aligned} C(S_0, 0; K, T) &= (S_0 - K)^+ + 1/2 E^Q \left[\int_0^T \delta(S_t - K) \sigma^2(S_t, t) dt \right] \\ &= (S_0 - K)^+ + 1/2 \int_0^T E^Q [\delta(S_t - K) \sigma^2(S_t, t)] dt \end{aligned}$$

- By differentiation wrt T on both sides:

$$C_T(S_0, 0; K, T) = E^Q [\delta(S_T - K) \sigma^2(S_T, T)] / 2$$

Proof of Dupire's formula

$$C_T(S_0, 0; K, T) = E^Q[\delta(S_T - K)\sigma^2(S_T, T)]/2$$

- Denote $\phi(x, t)$ the risk-neutral density at time t ; then:

$$E^Q[\delta(S_T - K)\sigma^2(S_T, T)] = \int \delta(x - K)\sigma^2(x, T)\phi(x, T)dx = \sigma^2(K, T)\phi(K, T)$$

by the defining property of the Dirac distribution

- Notice that the risk-neutral density is the price of a tight butterfly:

$$\phi(K, T) = C_{KK}(S_0, 0; K, T)$$

- Which concludes the proof:

$$C_T(S_0, 0; K, T) = \sigma^2(K, T)C_{KK}(S_0, 0; K, T)/2$$

Summary

- Hedging European calls monetizes convexity in return for time value decay:

$$Pnl = \theta \cdot \Delta t + \frac{1}{2} \cdot \Gamma \cdot (\Delta S)^2$$

- Modeling means assuming something about volatility $\sigma_t^2 = \frac{(dS_t)^2}{dt}$
 - ▶ Bachelier: constant volatility σ
 - ▶ Black and Scholes: proportional volatility $\sigma_L S_t$
 - ▶ Time-dependent models: $\sigma_N(t)$ or $\sigma_L(t) S_t$
 - ▶ Local volatility: $\sigma(S_t, t)$, including parametric specifications
- Many other, more sophisticated models exist but are not covered in this class
- The pricing PDE relates volatility model to pricing function such that PnL is 0 given volatility:

$$\frac{1}{2} \Gamma \hat{\sigma}^2 = -\theta$$

Summary

- The Feynman-Kac theorem expresses the solution of this PDE, given natural boundary $C(S_T, T) = (S_T - K)^+$, as the conditional expectation:

$$C(S_t, t; K, T) = E[(S_T - K)^+ | S_t]$$

under probabilistic dynamics:

$$dS_t = \sigma dW_t$$

- The only real-world feature of price series (S_t) that matters is volatility σ
In particular, the drift vanishes with delta-hedge: this is often called *risk-neutral pricing*
- Simple models Bachelier and Black and Scholes have closed-form solutions:

$$\text{Bach}(S_t, t; K, T; \hat{\sigma}) = (S_t - K)N(d) + \hat{\sigma}\sqrt{T-t}n(d)$$

$$\text{BS}(S_t, t; K, T; \hat{\sigma}) = S_t N(d_1) - KN(d_2)$$

- These formulas are used to compute prices given volatility, or by inversion, to *imply* volatility from prices

Summary

- With time-dependent models, implied volatility is a quadratic average of local:

$$\hat{\sigma}^2(T) = \frac{\int_0^T \sigma^2(t) dt}{T} \iff \sigma^2(t) = \hat{\sigma}^2(t) + t \hat{\sigma}^{\prime 2}(t)$$

- This remains true for full local volatility models, with gamma weighting:

$$\hat{\sigma}_N^2(K, T) = \frac{E^{Q(LV)}\{\int_0^T \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)] \sigma^2(S_t, t) dt\}}{E^{Q(LV)}\{\int_0^T \Gamma_B[S_t, t; K, T; \hat{\sigma}_N(K, T)] dt\}}$$

- The (much) more useful relation between prices and local volatilities is Dupire's dual PDE:

$$C_T = \frac{1}{2} \sigma^2 C_{kk}$$

Summary

- The break-even point analysis $Pnl = \theta \cdot \Delta t + \frac{1}{2} \cdot \Gamma \cdot (\Delta S)^2$ results in the mis-replication formula:
$$(S_T - K)^+ = C_M(S_0, 0) + \int_0^T \Delta_M(S_t, t) dS_t + \frac{1}{2} \int_0^T \Gamma_M(S_t, t) [(\sigma'_t)^2 - \sigma_M^2(S_t, t)] dt$$
- This formula, particularly the mis-replication term, is critical for:
 - ▶ Understanding how option trading plays out in the real world
 - ▶ Explaining PnL and managing option books accordingly
 - ▶ Elucidating and computing price differences between models
 - ▶ Appreciating what can legitimately cause the market implied volatility smile
 - ▶ Relating local and implied volatility
 - ▶ And even proving Dupire's celebrated formula!